The Language of Medicine 13th edition

Davi-Ellen Chabner

Chapter 19
Cancer Medicine (Oncology)

Chapter Goals (Slide 1 of 2)

- Identify medical terms that describe the growth and spread of tumors.
- Recognize terms related to the causes, diagnosis, and treatment of cancer.
- Review how tumors are classified and described by pathologists.

Chapter Goals (Slide 2 of 2)

- Describe x-ray studies, laboratory tests, and other procedures used by physicians for determining the presence and extent of spread (staging) of tumors.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 19
Lesson 19.1

Introduction

Cancer: characterized by unrestrained and excessive growth of cells

Malignant tumors: compress, invade, and destroy surrounding tissues

Cancer

- Cancer is responsible for 20% of all deaths in the United States.
- More than half of people who develop cancer are cured.

QUICK QUIZ (Slide 1 of 2)

1. **Which cancers are the most common causes of cancer death for women?**

   A. Lung, breast, colorectal
   B. Lung, colorectal, breast
   C. Breast, lung, colorectal
   D. Colorectal, breast, lung

QUICK QUIZ (Slide 2 of 2)

2. Which cancers are the most common causes of cancer death for men?

A. Lung, prostate, colorectal
B. Lung, colorectal, prostate
C. Prostate, lung, colorectal
D. Colorectal, prostate, lung

Carcinogenesis

- **Environmental agents**
  - ➢Chemical carcinogens
  - ➢Radiation
  - ➢Viruses (RNA and DNA)
  - ➢Oncogenes (ras/colon cancer, myc/lymphoma, and bcr-abl/chronic myelogenous leukemia)
- **Heredity**
  - ➢Retinoblastoma, polyposis coli

Genes Implicated in Hereditary Cancers

TABLE 19-2	EXAMPLES OF GENES IMPLICATED IN HEREDITARY CANCERS	
CANCER	**GENE**	**CHROMOSOMAL LOCATION***
Breast; ovarian	*BRCA1*	17q21
Breast; ovarian	*BRCA2*	3q12-13
Li-Fraumeni syndrome (multiple cancers)	*TP53*	17p13
Polyposis coli syndrome	*APC*	5q21
Renal cell carcinoma	*VHL*	3p21-26
Retinoblastoma	*RB1*	13q14
Wilms tumor	*WT1*	11p13

*The first number is the chromosome; p is the short arm of the chromosome, and q is the long arm of the chromosome. The second number is the region (band) of the chromosome.

Classification of Cancerous Tumors

Carcinomas: epithelial cell origin, 90% of all malignancies are carcinomas

Carcinomas and the Epithelial Tissues from Which They Derive

TABLE 19-3 CARCINOMAS AND THE EPITHELIAL TISSUES FROM WHICH THEY DERIVE

TYPE OF EPITHELIAL TISSUE	MALIGNANT TUMOR (CARCINOMAS)
Gastrointestinal tract	
Colon	Adeno**carcinoma** of the colon
Esophagus	Esophageal **carcinoma**
Liver	Hepatocellular **carcinoma** (hepatoma)
Stomach	Gastric adeno**carcinoma**
Glandular tissue	
Adrenal glands	**Carcinoma** of the adrenals (adrenocortical **carcinoma**)
Breast	**Carcinoma** of the breast
Pancreas	**Carcinoma** of the pancreas (pancreatic adeno**carcinoma**)
Prostate	**Carcinoma** of the prostate
Salivary glands	Adenoid cystic **carcinoma**
	Carcinoma of the thyroid
Thyroid	Renal cell **carcinoma** (hypernephroma)
Kidney and bladder	Transitional cell **carcinoma** of the bladder
Lung	Adeno**carcinoma** (bronchioloalveolar) Large cell **carcinoma** Small cell **carcinoma** Squamous cell **carcinoma** (epidermoid)
Reproductive organs	Adeno**carcinoma** of the uterus Squamous **carcinoma** of the penis Chorio**carcinoma** of the uterus or testes Cystadeno**carcinoma** (mucinous or serous) of the ovaries Seminoma and embryonal cell **carcinoma** (testes) Squamous cell (epidermoid) **carcinoma** of the vagina or cervix
Skin	
Basal cell layer	Basal cell **carcinoma**
Melanocyte	Malignant melanoma
Squamous cell layer	Squamous cell **carcinoma**

Classification of Cancerous Tumors (Slide 1 of 2)

Sarcomas: connective tissue origin, 5% of all malignancies

Sarcomas and the Connective Tissues from Which They Derive

TABLE 19-3	CARCINOMAS AND THE EPITHELIAL TISSUES FROM WHICH THEY DERIVE
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Gastrointestinal tract	
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Adrenal glands	**Carcinoma** of the adrenals (adrenocortical **carcinoma**)
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Pancreas	**Carcinoma** of the pancreas (pancreatic adeno**carcinoma**)
Prostate	**Carcinoma** of the prostate
Salivary glands	Adenoid cystic **carcinoma**
Thyroid	**Carcinoma** of the thyroid
Kidney and bladder	
	Renal cell **carcinoma** (hypernephroma) Transitional cell **carcinoma** of the bladder
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Skin	
Basal cell layer	Basal cell **carcinoma**
Melanocyte	Malignant melanoma
Squamous cell layer	Squamous cell **carcinoma**

Classification of Cancerous Tumors (Slide 2 of 2)

Mixed tissue tumors: tissues capable of differentiating into epithelial and connective tissue

TABLE 19-5	MIXED-TISSUE TUMORS
TYPE OF TISSUE	**MALIGNANT TUMOR**
Kidney	Wilms tumor (embryonal adenosarcoma)
Ovaries and testes	Teratoma (tumor composed of bone, muscle, skin, gland cells, cartilage, etc.) Germ cell tumor

Chapter 19

Lesson 19.2

Pathological Descriptions Gross

- Cystic
- Fungating
- Inflammatory
- Medullary
- Necrotic
- Polypoid
- Ulcerating
- Verrucous

Pathological Descriptions Microscopic

- Alveolar
- Carcinoma in situ
- Diffuse
- Dysplastic
- Epidermoid
- Follicular
- Nodular
- Papillary
- Pleomorphic
- Scirrhous
- Undifferentiated

Grading and Staging of Tumors

Grade: degree of maturity or differentiation under the microscope

Stage: extent of spread in the body

International TNM Staging System for Lung Cancer

Regional Lymph Nodes (N)	
N0	No node involvement
N1	Ipsilateral (same side as for the primary tumor) bronchopulmonary or hilar nodes involved
N2	Ipsilateral mediastinal nodes or ligament involved
N3	Contralateral mediastinal, hilar nodes, or supraclavicular (collarbone) nodes involved
Distant Metastasis (M)	
M0	No metastasis
M1	Metastases present with site specified (e.g., brain, liver)

TNM, tumor-node-metastasis. <, less than; >, more than.

Chapter 19

Lesson 19.3

Cancer Treatment: Surgery (Slide 1 of 2)

- **Debulking procedure**: may be used to remove as much of primary tumor mass as possible

- **Adjuvant** (assisting) radiation therapy and/or chemotherapy: after removal of primary tumor to prevent recurrence at distant sites

Cancer Treatment: Surgery (Slide 2 of 2)

- Cauterization
- Core needle biopsy
- Cryosurgery
- En bloc resection
- Excisional biopsy
- Exenteration
- Fine needle aspiration biopsy
- Fulguration
- Incisional biopsy

Cancer Treatment: Radiation Therapy (Radiation Oncology)

- Brachytherapy
- Electron beams
- External beam radiation (teletherapy)
- Fields
- Fractionation
- Gray (Gy)
- Linear accelerator
- Photon therapy
- Proton therapy
- Radiocurable tumor
- Radioresistant tumor
- Radiosensitive tumor
- Radiosensitizers
- Simulation
- Stereotactic radiosurgery

Radiation Therapy Side Effects

- Alopecia (baldness)
- Fibrosis (increase in connective tissue)
- Infertility (inability to contribute to conception of a child)
- Mucositis (inflammation and ulceration of mucous membranes)
- Myelosuppression (bone marrow depression)
- Nausea and vomiting
- Pneumonitis
- Secondary tumors (new types of tumors)
- Xerostomia (dryness of mouth)

Chapter 19
Lesson 19.4

Chemotherapy, Biological Therapy, and Differentiating Agents

- Alkylating agents
- Antibiotics
- Antimetabolites
- Antimitotics
- Hormonal agents

Cancer Chemotherapeutic Agents and the Cancers They Treat (slide 1 of 2)

TABLE 19-7 SELECTED CANCER CHEMOTHERAPEUTIC AND HORMONAL AGENTS AND THE CANCERS THEY TREAT

CHEMOTHERAPEUTIC AGENT	TYPE OF CANCER
Alkylating Agents	
Cytoxan (cyclophosphamide)	Lymphoma
Platinol (cisplatin*)	Testicular, ovarian, lung
Antibiotics	
Adriamycin (doxorubicin)	Breast
Blenoxane (bleomycin)	Testicular, Hodgkin lymphoma
Onivyde (irinotecan)	Colorectal
Antimetabolites	
Alimta (pemetrexed)	Lung
Ara-C (cytarabine)	AML
Folex (methotrexate, MXT)	Acute lymphoid leukemia (ALL)
Antimitotics	
Oncovin (vincristine)	Lymphoma
Taxol (paclitaxel)	Breast, ovarian, lung

Cancer Chemotherapeutic Agents and the Cancers They Treat (slide 2 of 2)

TABLE 19-7 SELECTED CANCER CHEMOTHERAPEUTIC AND HORMONAL AGENTS AND THE CANCERS THEY TREAT

HORMONAL AGENT	TYPE OF CANCER
Hormones and Hormone Antagonists	
Arimidex (anastrozole)	Breast
Lupron (leuprolide)	Prostate
Nolvadex (tamoxifen)	Breast
Zytiga (abiraterone)	Prostate

Note: Generic names are in parentheses. Often drugs are given in combinations according to carefully planned regimens called **protocols.**
*Acts in a similar manner to alkylating agents

Vocabulary (Slide 1 of 18)

Term	Meaning/Definition
adjuvant chemotherapy	Assisting primary treatment
alkylating agents	Synthetic chemicals containing alkyl groups that attack DNA
anaplasia	Loss of differentiation of cells
angiogenesis	Process of forming new blood vessels

Vocabulary (Slide 2 of 18)

Term	Meaning/Definition
antibiotics	Chemicals produced by bacteria or primitive plants; inhibit growth of cells
antimetabolites	Chemicals that prevent cell division; inhibit formation of substances needed to make DNA
antimitotics	Drugs that block cell division
apoptosis	Programmed cell death

Vocabulary (Slide 3 of 18)

Term	Meaning/Definition
benign tumor	Noncancerous growth (neoplasm)
biological response modifiers	Produced by normal cells; directly block tumor growth or stimulate immune system to fight cancer
biological therapy	Use of the body's own defenses to destroy tumor cells
brachytherapy	Use of radiation placed directly on or within the cancer

Vocabulary (Slide 4 of 18)

Term	Meaning/Definition
carcinogens	Agents that cause cancer
carcinoma	Cancerous tumor made up of cells of epithelial origin
cellular oncogenes	Pieces of DNA, activated by mutations or dislocation, that can cause a normal cell to become malignant
chemotherapy	Treatment with drugs

Vocabulary (Slide 5 of 18)

Term	Meaning/Definition
combination chemotherapy	Use of several chemotherapeutic agents together in treatment of tumors
dedifferentiation	Loss of differentiation of cells
deoxyribonucleic acid (DNA)	Genetic material within the nucleus of a cell; controls cell division and protein synthesis
differentiating agents	Drugs that promote tumor cells to differentiate, stop growing, and die

Vocabulary (Slide 6 of 18)

Term	Meaning/Definition
differentiation	Specialization of cells
electron beams	Low-energy beams of radiation for treatment of skin or surface tumors
encapsulated	Surrounded by a capsule; benign tumors are encapsulated
external beam irradiation	Applying radiation to a tumor from a source outside the body

Vocabulary (Slide 7 of 18)

Term	Meaning/Definition
fields	Dimensions of body areas undergoing irradiation
fractionation	Giving radiation in small, repeated doses
genetic screening	Testing family members to determine if they have inherited a cancer-causing gene
grading tumors	Evaluating the degree of maturity of tumor cells

Vocabulary (Slide 8 of 18)

Term	Meaning/Definition
gray (gy)	Unit of absorbed radiation dose
gross description of tumors	Visual appearance of tumors to the naked eye
infiltrative	Extending beyond normal tissue boundaries into adjacent tissues
invasive	Having the ability to enter and destroy surrounding tissue

Vocabulary (Slide 9 of 18)

Term	Meaning/Definition
irradiation	Exposure to any form of radiant energy such as light, heat, or x-rays
linear accelerator	Large electronic device that produces high-energy x-ray beams for treatment of deep-seated tumors
malignant tumor	Tumor having the characteristics of continuous growth, invasiveness, and metastasis
mesenchymal	Embryonic connective tissue

Vocabulary (Slide 10 of 18)

Term	Meaning/Definition
metastasis	Spread of a malignant tumor to a secondary site
microscopic description of tumors	Appearance of tumors when viewed under a microscope
mitosis	Replication of cells
mixed-tissue tumors	Tumors composed of different types of tissue

Vocabulary (Slide 11 of 18)

Term	Meaning/Definition
modality	Method of treatment, such as surgery, chemotherapy, or irradiation
molecularly targeted therapy	Anticancer drugs designed to block the function of growth factors, their receptors, and signaling pathways in specific tumor cells
morbidity	Condition of being unwell; deficient in normal function
mucinous	Containing mucus

Vocabulary (Slide 12 of 18)

Term	Meaning/Definition
mutation	Change in DNA; may be spontaneous or caused by chemicals, radiation, or viruses
neoplasm	New growth; benign or malignant tumors
nucleotide	Unit of DNA composed of a sugar, phosphate, and a base
oncogene	Region of DNA in tumor cells or in viruses that causes cancer

Vocabulary (Slide 13 of 18)

Term	Meaning/Definition
palliative	Relieving, but not curing symptoms
pedunculated	Possessing a stem or stalk
photon therapy	Radiation therapy using energy in the form of x-rays or gamma rays
protocol	Detailed plan for treatment of an illness

Vocabulary (Slide 14 of 18)

Term	Meaning/Definition
proton therapy	Use of protons produced by a cyclotron to deposit a dose of radiation at a tightly focused point in the body
radiation	Energy carried by a stream of particles
radiocurable tumor	Tumor that is destroyed by radiation therapy
radioresistant tumor	Tumor that requires large doses of radiation to be destroyed

Vocabulary (Slide 15 of 18)

Term	Meaning/Definition
radiosensitive tumor	Tumor in which radiation can cause the death of cells without serious damage to surrounding tissue
radiosensitizers	Drugs that increase the sensitivity of tumors to x-rays
radiotherapy	Treatment of tumors using doses or radiation
relapse	Recurrence of tumor after treatment

Vocabulary (Slide 16 of 18)

Term	Meaning/Definition
remission	Partial or complete disappearance of symptoms of disease
ribonucleis acid (RNA)	Cellular substance that represents a copy of DNA; directs formation of new protein inside cells
sarcoma	Cancerous tumor derived from connective or flesh tissue
serous	Having the appearance of a thin, watery fluid
sessile	Having no stem

Vocabulary (Slide 17 of 18)

Term	Meaning/Definition
simulation	Study using CT scan or MRI to map treatment before RT is g
solid tumor	Tumor composed of a mass of cells
staging of tumors	System of evaluating the extent of spread of tumors
stereotactic radiosurgery	Delivery of dose of radiation under stereotactic guidance

Vocabulary (Slide 18 of 18)

Term	Meaning/Definition
steroids	Complex, naturally occurring chemicals derived from cholesterol
surgical procedures to treat cancer	Methods of removing cancerous tissue
viral oncogenes	Pieces of DNA from viruses that infect a normal cell and cause it to become malignant
virus	Infectious agent that reproduces by entering a host cell and using the host's genetic material to copy itself

Clinical Procedures to Detect or Treat Malignancies

- Bone marrow biopsy
- Bone marrow or stem cell transplant
- CT scans
- Fiberoptic colonoscopy
- Exfoliative cytology
- Laparoscopy
- Mammography
- MRI
- Needle biopsy
- Radionuclide scans
- Ultrasound
- X-rays

Abbreviations (Slide 1 of 4)

Abbreviation	Meaning
AFP	Alpha-fetoprotein
ATRA	All-transretinoic acid
BMT	Bone marrow transplantation
BRC	breakpoint cluster region
bx	Biopsy
CA	Cancer
CEA	Carcinoembryonic antigen
cGy	Centigray (one hundredth of a gray) or rad
chemo	Chemotherapy
CR	Complete response – disappearance of all tumor
CSF	Colony-stimulating factor
DES	Diethylstilbestrol
DNA	Deoxyribonucleic acid

Abbreviations (Slide 2 of 4)

Abbreviation	Meaning
EGFR	Epidermal growth factor receptor
ER	Estrogen receptor
EPO	Erythropoietin
FNA	Fine needle aspiration
5-FU	5-fluorouracil
Ga	Gallium
GIST	Gastrointestinal stromal tumor
Gy	Gray
H & E	Hematoxylin and eosin
HER2-neu	Growth factor gene highly activated on certain types of breast cancer
HNPCC	Hereditary non-polyposis

Abbreviations (Slide 3 of 4)

Abbreviation	Meaning
IGRT	Intensity-modulated gated radiation therapy
IHC	Immunohistochemistry
IMRT	Intensity-modulated radiation therapy
IORT	Intraoperative radiation therapy
Mets	Metastases
MoAb	Monoclonal antibody
NED	No evidence of disease
NF	Neurofirbromatosis
NHL	Non–Hodgkin lymphoma
NSCLC	Non-small cell lung cancer
Pap smear	Papanicolaou smear
PD	Progressive disease – tumor increases in size

Abbreviations (Slide 4 of 4)

Abbreviation	Meaning
PR	Partial response – tumor one half its original size
prot	Protocol
PSA	Prostate-specific antigen
PSCT	Peripheral stem cell transplantation
PSRS	Proton stereotactic radiosurgery
RNA	Ribonucleic acid
RT	Radiation therapy
SD	Stable disease – tumor does not shrink or grow
TNM	Tumor, nodes, metastases
VEGF	Vascular endothelial growth factor
XRT, RT	Radiation therapy